Part I: Theory & Definitions

(Chapter-wise Collection of Recurring Theory Questions)

Module 1: Properties of Matter

1.1 Important Definitions

1. Define: Stress, Strain, and Young's Modulus. (W-24, S-25)
2. Define: Ductility, Brittleness, and Elasticity. (S-23)
3. Define: Elastic Limit, Yield Point. (S-22, W-24)

1.2 Theory Questions

1. **Stress-Strain Diagram:** Draw the Stress-Strain diagram for a ductile material (like copper or steel) with necessary notations. Explain terms like Elastic Limit and Upper Yield Point. (S-25, W-24, S-22, W-18)
2. **Cantilever:** What is a Cantilever? Derive the expression for the depression at the free end of a thin beam clamped horizontally at one end and loaded at the other end. (W-24, S-23, W-23)
3. **Torsional Pendulum:** What is a torsional pendulum? Derive the expression for the time period of vibration for a torsional pendulum. Derive the equation for twisting couple per unit twist of a wire. (S-25, S-24, W-24)
4. **Elastic Hysteresis:** Explain: (i) Elastic hysteresis, (ii) Elastic after-effect, and (iii) Elastic fatigue. (S-25)
5. **I-Shaped Girders:** Write a short note on I-shaped girders and their applications. (S-23)

Module 2: Waves, Motion & Acoustics

2.1 Important Definitions

1. Define: Simple Harmonic Motion (SHM), Free Vibration, Forced Vibration, and Resonance. (S-25, W-24)
2. Define: Transverse waves, Longitudinal waves, Ultrasonic waves. (S-25, W-24)
3. Define: Reverberation, Reverberation Time, Absorption Coefficient, Echo, Echelon effect. (S-25, W-24, S-23)

2.2 Theory Questions

1. **SHM:** Derive the differential equation of simple harmonic motion. Obtain the equation for kinetic energy and potential energy of a simple harmonic oscillator. Show that total energy remains constant. (W-24, S-25, W-22)
2. **Damped Harmonic Motion:** What is damped harmonic motion? Derive the differential equation and general solution for damped harmonic motion. (S-25, W-19, W-23)
3. **Ultrasonic Production:** Explain the construction and working of the **Piezoelectric method** for the production of ultrasonic waves with a suitable diagram. List its merits and demerits. (S-25, W-24, S-23, S-24)
4. **Acoustics of Buildings:** Discuss various factors affecting the acoustics of a building (auditorium) and give their remedies. (S-25, W-24, S-24)
5. **Reverberation Time:** Write Sabine's formula for reverberation time and explain each term. (S-24, W-19)
6. **NDT:** What is Non-destructive testing (NDT)? Discuss the use of ultrasound in NDT (Ultrasonic Flaw Detector/Pulse Echo System). (S-25, S-23, W-22)
7. **Detection:** Explain the method for the detection of ultrasonic waves (Kundts tube method). (S-25 Syllabus)

Module 3: Optics

3.1 Important Definitions

1. Define: Interference of light, Diffraction of light. (S-25, W-24)
2. State: Huygens' Principle. (S-25)
3. Differentiate: Constructive and Destructive interference. (W-24)
4. Differentiate: Interference and Diffraction. (W-24)

3.2 Theory Questions

1. **Young's Double Slit:** Explain Young's double slit experiment. Discuss the conditions for constructive and destructive interference in terms of path difference. (S-25)
2. **Newton's Rings:** Explain the formation of Newton's rings with a suitable diagram. Derive the equation for the radius of bright and dark rings produced by the interference of reflected light. (S-25)
3. **Michelson Interferometer:** Describe the principle, construction, and working of a Michelson interferometer. (W-24)
4. **Diffraction:** Explain Fresnel and Fraunhofer diffraction. Discuss the resolving power of a grating. (S-25)
5. **Superposition:** State and explain the principle of superposition of waves. (W-24)
6. **Anti-Reflection Coating:** Write a note on Anti-reflection coating. (Syllabus Topic)

Module 4: Quantum Physics

4.1 Theory Questions

1. **De Broglie Waves:** What are De-Broglie waves (Matter waves)? Write the equation of De-Broglie wavelength. (S-25)
2. **Heisenberg's Uncertainty:** Explain Heisenberg's uncertainty principle and write it in terms of position and momentum. (S-25, W-24)
3. **Schrodinger's Equation:** Derive/Write the time-independent Schrödinger's wave equation and explain each term. (S-25, W-24)
4. **Black Body Radiation:** Write a short note on black body radiation. (W-24)
5. **STM:** Write a short note on the Scanning Tunneling Microscope (STM). (S-25, W-24)
6. **Particle in a Box:** Discuss the particle in a one-dimensional rigid box. (Syllabus Topic)

Module 5: Lasers

5.1 Important Definitions

1. Define: Population Inversion, Metastable State, Pumping, Active Medium. (S-25, W-24)
2. Define: Spontaneous Emission, Stimulated Emission, Absorption. (S-25, W-24, S-24)

5.2 Theory Questions

1. **Einstein's Coefficients:** What are Einstein's coefficients? Obtain the relation between Einstein's coefficients (A and B). (S-25, W-24, S-22)
2. **Properties & Applications:** Write the properties of LASER. State applications of lasers in science, engineering, and medicine. (S-25, W-24, S-22)
3. **Ruby Laser:** Discuss the construction and working of the Ruby laser with a suitable energy level diagram. (S-25, S-24, W-23)
4. **He-Ne Laser:** Discuss the construction and working of the He-Ne gas laser with a suitable diagram. (S-25, S-24, S-23)
5. **Population Inversion:** Explain the concept of population inversion and why it is necessary for the production of a laser. (S-25)

Module 6: New Engineering Materials

6.1 Important Definitions

1. Define: Superconductivity, Critical Temperature (T_c), Critical Magnetic Field (H_c), Critical Current Density (J_c). (W-24)
2. Define: Nanomaterials. (S-25, W-24)

6.2 Theory Questions

1. **Superconductors:**
   - Discuss the properties of superconductors (Meissner effect, Isotope effect). (S-25, W-24)
   - Differentiate between Type-I and Type-II superconductors. (S-25, S-24, W-23)
   - Explain the BCS theory of superconductivity. (S-25, W-24)
   - Write a note on Josephson Junction and SQUID. (S-25, W-22)
   - State applications of superconductors. (S-25, W-24)
2. **Semiconductors:**
   - Differentiate between N-type and P-type semiconductors. (S-25, W-24)
   - Explain the working of the P-N junction diode and its I-V characteristics. (W-24, Syllabus)
   - Explain Zener diode and its characteristics. (Syllabus Topic)
3. **Nanomaterials:**
   - Explain the synthesis of nanomaterials using the **Ball Milling** method. (S-25)
   - Explain the **PVD** (Physical Vapor Deposition) method. (Syllabus Topic)
   - Write the applications of nanomaterials. (W-24)

Part II: Numerical Problems

(Chapter-wise Collection of All Numericals from Past Papers)

Module 1: Properties of Matter (Numericals)

1. **(S-25)** Find the force required to increase the length of a steel wire of $10^{-6}m^2$ area of cross-section by 50% whose Young's modulus is $2 \times 10^{11} N/m^2$.
2. **(W-24)** An unknown weight is attached to the lower end of a wire of length 4 m, radius 0.7 mm, extends it by 0.8 mm. If $Y = 2 \times 10^{11} N/m^2$, find the unknown weight.
3. **(S-23)** Calculate Poisson's ratio and the rigidity modulus of copper using the following data: Young's modulus $10.5 \times 10^{10} Nm^{-2}$ and the bulk modulus of copper $14.3 \times 10^{10} Nm^{-2}$.
4. **(W-22)** A wire of length 2 m and radius 0.5 mm is clamped at one of its ends. Calculate the torque required to twist the other end by $90°$. The rigidity modulus of the wire is $2.8 \times 10^{10} N/m^2$.

Module 2: Waves, Motion & Acoustics (Numericals)

1. **(S-25)** A particle of mass 0.50 kg executes a simple harmonic motion. If it crosses the center of oscillation with a speed of $10m/s$, find the amplitude of the motion. (force constant $k = 50N/m$).
2. **(S-25)** The volume of a room is $1500m^3$. The wall area of the room is $250m^2$, the floor and the ceiling area is $150m^2$ each. The average sound absorption coefficient for the wall, floor and the ceiling are 0.25, 0.06 and 0.80, respectively. Calculate the average absorption coefficient and the reverberation time.
3. **(W-24)** The volume of a hall is 475 m^3. The area of wall is $200m^2$, area of floor and ceiling each is $100m^2$. If the absorption co-efficient of wall, ceiling and floor are 0.025, 0.02 and 0.55, respectively. Calculate the reverberation time for the hall.
4. **(W-24)** An ultrasonic source of 0.09 MHz sends down a pulse towards the seabed which returns after 0.55 sec. The velocity of sound in water is $1800m/s$. Calculate the depth of the sea and wavelength of pulse.
5. **(S-24)** An ultrasonic wave of 7.2 MHz sends down a pulse towards the seabed, which returns after 1.2 s. The velocity of sound in seawater is $1800m/s$. Calculate the depth of sea and wavelength of pulse.

Module 3: Optics (Numericals)

1. **(S-25)** In a Young's double slit experiment, the two slits which are 0.5 mm apart are illuminated by the yellow light of wavelength 589.0 nm. Calculate the distance between two

consecutive bright bands in the interference pattern observed on a screen 2 m away from the slits.

2. **(W-24)** Calculate the minimum number of lines in a grating which will just resolve the lines of wavelength 5890 Å and 5896 Å in the second order.

Module 4: Quantum Physics (Numericals)

1. **(S-25)** Calculate de-Broglie wavelength associated with (i) A man of 50 kg mass walking with the speed of $2m/s$, and (ii) An electron moving with the speed of $3 \times 10^5 m/s$. ($h = 6.63 \times 10^{-34} J - s$, electron mass $= 9.11 \times 10^{-31}$ kg).

Module 5: Lasers (Numericals)

1. **(S-25)** In He-Ne gas laser, Ne atoms emit laser photons of wavelength 632.8 nm while electrons make transitions from E_3 state to E_2 state. Find the energy difference between these two states.

2. **(S-25)** A ruby laser emits light of wavelength 694.3 nm. Calculate the frequency and the energy of the laser photons emitted. ($h = 6.63 \times 10^{-34} J - s, c = 3 \times 10^8 m/s$).

3. **(W-23)** In He-Ne laser system, the two energy levels of Ne involved in lasing action have energy values of 20.66 eV and 18.70 eV. What will be the wavelength of a laser beam produced?

Module 6: New Engineering Materials (Numericals)

1. **(S-25)** Transition temperature of mercury of mass number 202 is 4.20 K. Find the transition temperature of isotope of mercury of mass number 204.

2. **(S-25)** Calculate the critical current for a superconducting wire having radius of 0.2 mm at 5 K. Critical temperature of material of wire is 7.2 K and $H_c(0)$ is $2.2 \times 10^4 A/m$.

3. **(W-24)** The critical temperature for a metal with isotopic mass 199.5 is 4.185 K. Calculate the isotopic mass if the critical temperature falls to 4.133 K.

4. **(S-24)** The critical current passing through a superconducting wire of radius 5 mm is 21 A. Calculate critical magnetic field.

5. **(S-23)** Calculate the critical current for a wire of Pb having a diameter of 1 mm at 4.5 K. The critical temperature for Pb is 7.2 K and $H_c(0) = 6.5 \times 10^4 A/m$.